# DATA-DRIVEN E-COMMERCE PURCHASE INTENTION ANALYSIS USING PYTHON

Python Data Analytics Project Documentation

Dataset: Online Shoppers Purchasing Intention Dataset
Analysis environment: Google Colab

## 1. Project Overview

This project analyzes online shopper browsing sessions to understand how website engagement and session characteristics relate to purchase outcomes. Using Python, the notebook performs initial data exploration, duplicate removal, data standardization, feature engineering, descriptive statistical analysis, correlation analysis and visual exploration.

The analysis focuses on session-level behavior rather than unique customer-level behavior. The target variable, Revenue, indicates whether a session resulted in a purchase.

## 2. Data Source

| Item | Details |
|---|---|
| **Dataset name** | Online Shoppers Purchasing Intention Dataset |
| **Original source** | UCI Machine Learning Repository |
| **Dataset citation** | Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository. DOI: 10.24432/C5F88Y. |
| **Source URL** | https://archive.ics.uci.edu/dataset/468/online%2Bshoppers%2Bpurchasing%2Bintention%2Bdatase |
| **Project repository** | https://github.com/Bhavya-Chellapandian/online-shoppers-purchasing-intention-dataset |
| **Notebook input** | CSV loaded from the project's GitHub raw file URL |
| **Original size** | 12,330 session records and 18 attributes |
| **Cleaned size** | 12,205 session records and 18 original attributes; 8 engineered features added later |
| **Domain** | Retail / E-commerce / Online customer behaviour |
| **Data unit** | Website browsing session |

## 3. Problem Statement

E-commerce websites collect session-level information about page visits, time spent browsing, traffic sources and visitor types. However, these records do not automatically explain which browsing patterns are associated with completed purchases. This project explores the relationship between session

engagement and the Revenue outcome, and compares conversion across visitor type, month, weekend status, traffic type and region.

## 4. Objectives

1. Understand online shoppers browsing behaviour using session-level data.

2. Explore how administrative, informational, and product-related page visits and time spent relate to purchase outcomes.

3. Examine the distribution of purchasing and non-purchasing sessions using the Revenue target.

4. Compare session characteristics across visitor type, month, weekend status, traffic type, and other categorical attributes.

5. Identify data quality issues and useful patterns to investigate in the next stages of cleaning, statistical analysis, and visualization.

6. Translate the findings into practical recommendations for improving e-commerce conversion.

## 5. Attribute (Column / Feature) Details

| Attribute | Description |
|---|---|
| **Administrative** | Number of administrative pages visited in a session. |
| **Administrative_Duration** | Time spent on administrative pages. |
| **Informational** | Number of informational pages visited. |
| **Informational_Duration** | Time spent on informational pages. |
| **ProductRelated** | Number of product-related pages visited. |
| **ProductRelated_Duration** | Time spent on product-related pages. |
| **BounceRates** | Bounce rate measure associated with the session. |
| **ExitRates** | Exit rate measure associated with the session. |
| **PageValues** | Page-value measure associated with pages viewed before a transaction. |
| **SpecialDay** | Closeness of the session date to a special day. |
| **Month** | Month in which the session occurred. |
| **OperatingSystems** | Encoded operating-system category. |
| **Browser** | Encoded browser category. |
| **Region** | Encoded geographic region category. |
| **TrafficType** | Encoded traffic-source category. |
| **VisitorType** | Visitor category: returning, new or other. |

| | |
|---|---|
| **Weekend** | Boolean indicator of whether the session occurred on a weekend. |
| **Revenue** | Boolean target indicating whether the session resulted in a purchase. |

## 6. Tools & Technologies

- Python
- Pandas – data loading, cleaning, transformation and aggregation
- NumPy – numerical operations and feature engineering
- Matplotlib – chart creation
- Seaborn – statistical visualization and correlation heatmap
- Google Colab – execution environment

## 7. Data Pre-Processing and Feature Engineering

### 7.1 Data cleaning

- Created a working copy of the original DataFrame.
- Checked the dataset dimensions, column names, data types and non-null counts.
- Checked for missing values: none were reported across the 18 original columns.
- Identified 125 exact duplicate rows and removed them.
- Standardized column names to lowercase snake_case.
- Trimmed whitespace from text-category values.
- Ensured Revenue was represented as a Boolean target.
- Checked BounceRates and ExitRates ranges; both ranged from 0.000 to 0.200.

After duplicate removal, the cleaned dataset contained 12,205 rows and 18 columns. Since no missing values were present, missing-value imputation was not required.

### 7.2 Feature engineering

| Derived feature | Definition / purpose |
|---|---|
| **total_page_views** | Sum of administrative, informational and product-related page counts. |
| **total_page_time** | Sum of time spent across the three page categories. |
| **avg_time_per_page** | Total page time divided by total page views; zero-page sessions are set to missing for the division. |

| | |
|---|---|
| **administrative_view_share** | Administrative page views divided by total page views. |
| **informational_view_share** | Informational page views divided by total page views. |
| **product_related_view_share** | Product-related page views divided by total page views. |
| **browsed_products** | Boolean indicator: True when product_related is greater than zero. |
| **purchase_status** | Readable label: Purchase or No purchase, derived from revenue. |

## 8. Analysis and Visualizations

The notebook uses session-level exploratory analysis to compare engagement and conversion outcomes. The executed visualizations include a Spearman correlation heatmap, a comparison of median time spent by page category and purchase outcome, a purchase-outcome distribution, conversion by visitor type, conversion by month, and conversion comparisons by weekend, traffic type and region.

## 9. Exploratory Data Analysis (EDA)

- Initial dataset shape: 12,330 rows × 18 columns.
- All 18 columns had 12,330 non-null values; no missing values were detected.
- Data types: 7 integer, 7 floating-point, 2 object and 2 Boolean columns.
- Numerical and categorical/Boolean columns were separated for inspection.
- Exact duplicate rows: 125.
- After duplicate removal: 12,205 rows × 18 columns.
- Most common visitor type in the original dataset: Returning_Visitor (10,551 sessions).
- Most common month in the original dataset: May (3,364 sessions).

## 10. Statistical Analysis

Descriptive statistics were calculated for the cleaned dataset. Selected values are shown below.

| Measure | Mean | Median | Std. dev. | Minimum | Maximum |
|---|---|---|---|---|---|
| **Administrative page views** | 2.34 | 1 | 3.33 | 0 | 27 |
| **Informational page views** | 0.51 | 0 | 1.28 | 0 | 24 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Product-related page views** | | 32.05 | 18 | 44.59 | 0 | 705 |
| **Total page views** | | 34.89 | 20 | 46.63 | 0 | 746 |
| **Total page time** | | 1,323.45 | 690.96 | 2,043.87 | 0 | 69,921.65 |
| **PageValues** | | 5.95 | 0 | 18.65 | 0 | 361.76 |
| **BounceRates** | | 0.02 | 0.00 | 0.05 | 0 | 0.20 |
| **ExitRates** | | 0.04 | 0.02 | 0.05 | 0 | 0.20 |

The notebook also calculates conversion rates by visitor type, month, weekend, traffic type, region and page-value group. Spearman rank correlation is used to examine relationships among browsing measures.

## 11. Data Visualization

### 11.1 Univariate / outcome distribution

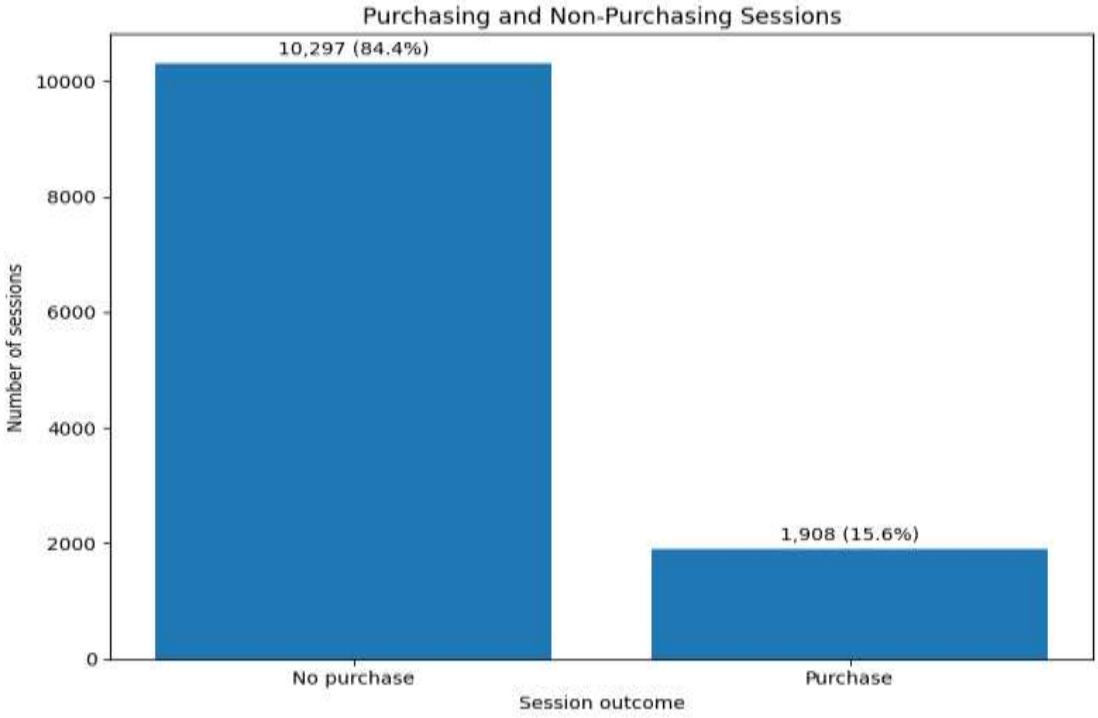



*Figure 1. Number of purchasing and non-purchasing sessions*

Interpretation: 1,908 of 12,205 sessions (15.63%) resulted in a purchase; 10,297 sessions (84.37%) did not. Non-purchasing sessions were approximately 5.4 times as numerous as purchasing sessions.

### 11.2 Bivariate analysis

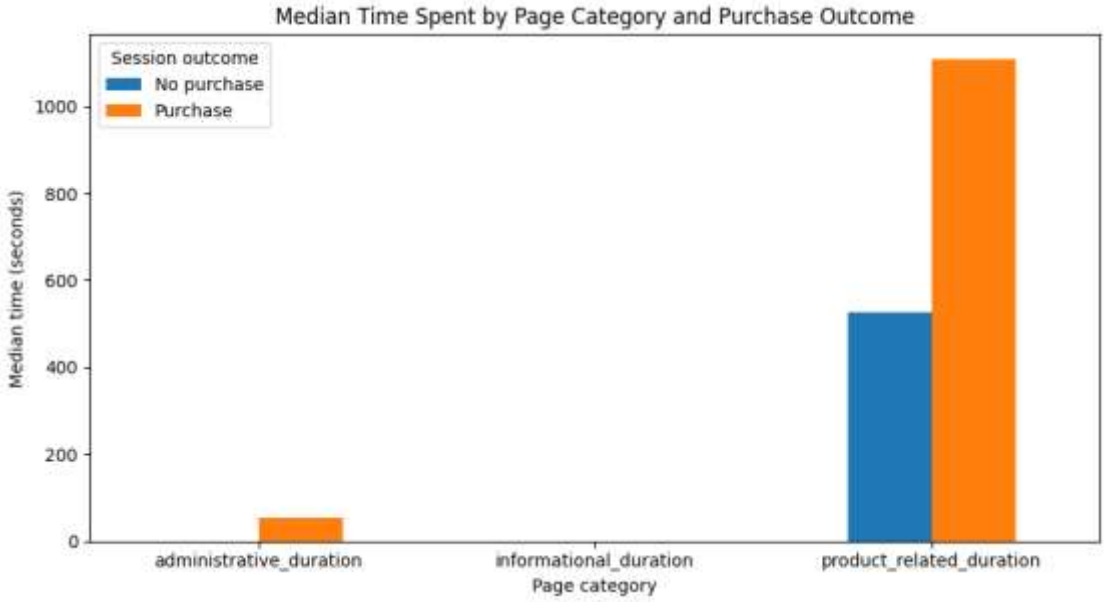



*Figure 2. Median time spent by page category and purchase outcome*

Interpretation: Purchasing sessions had a median product-related page time of 1,109.91 seconds, compared with 526.00 seconds for non-purchasing sessions. Median administrative-page time was 52.37 seconds for purchasing sessions and 0 seconds for non-purchasing sessions. Median informational-page time was 0 seconds in both groups.

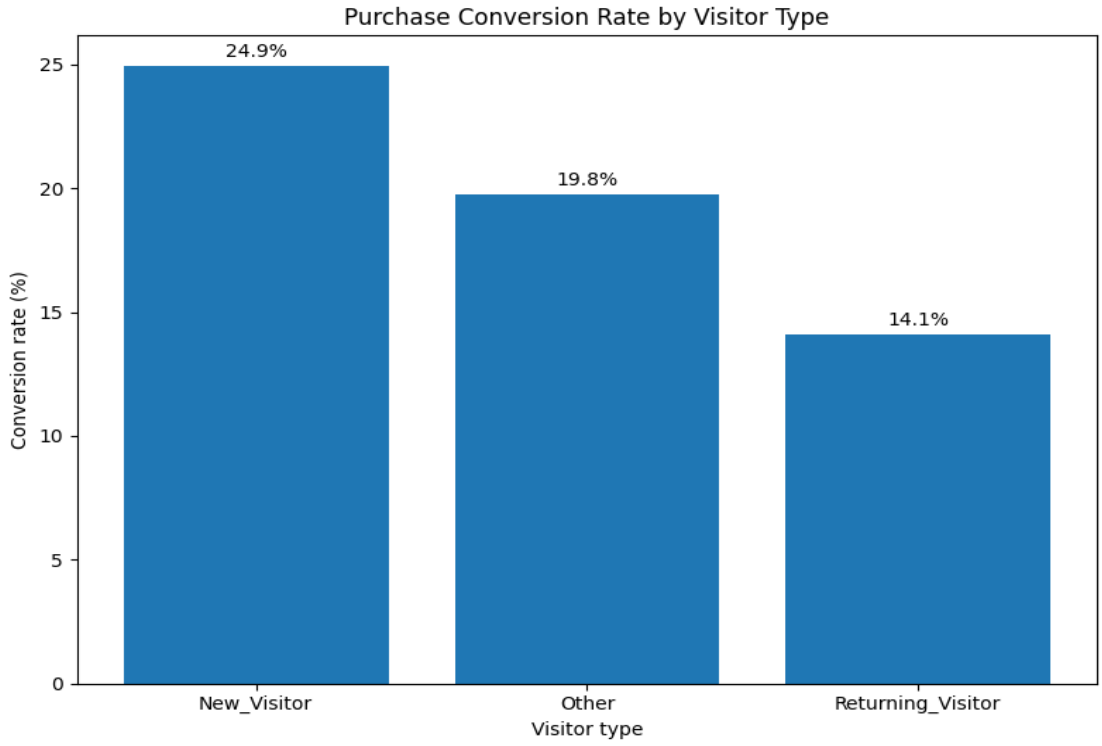



*Figure 3. Purchase conversion rate by visitor type*

Interpretation: New visitors had the highest conversion rate (24.93%; 422 of 1,693 sessions), followed by Other visitors (19.75%; 16 of 81) and Returning visitors (14.09%; 1,470 of 10,431). Returning visitors formed the largest session segment.

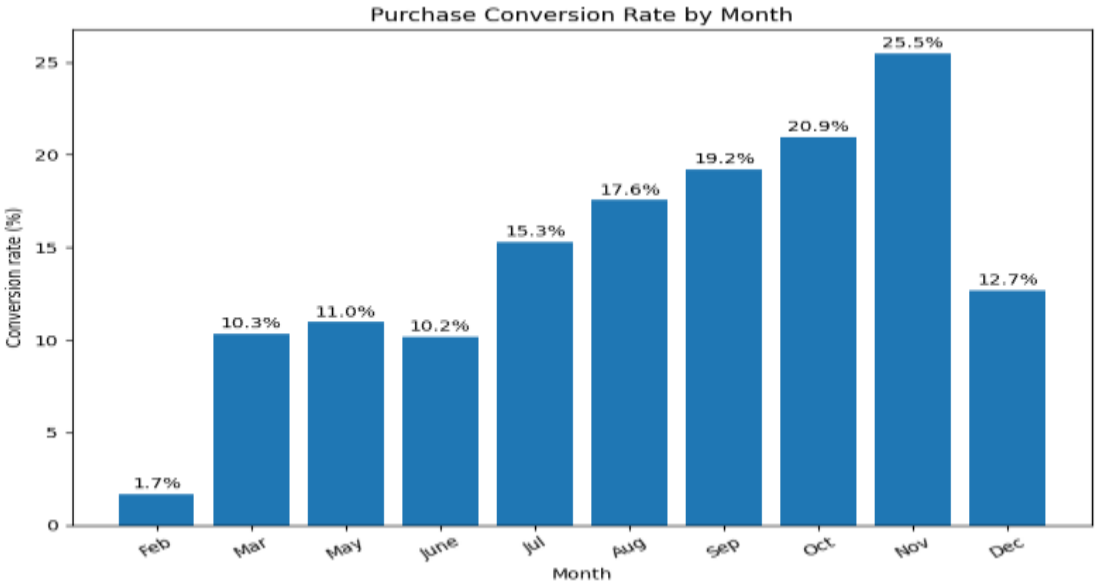



*Figure 4. Purchase conversion rate by month*

Interpretation: November had the highest observed monthly conversion rate (25.49%; 760 of 2,982 sessions), while February had the lowest (1.66%; 3 of 181). Monthly session volumes vary, so rates should be interpreted alongside counts. January and April are not represented in the cleaned results.

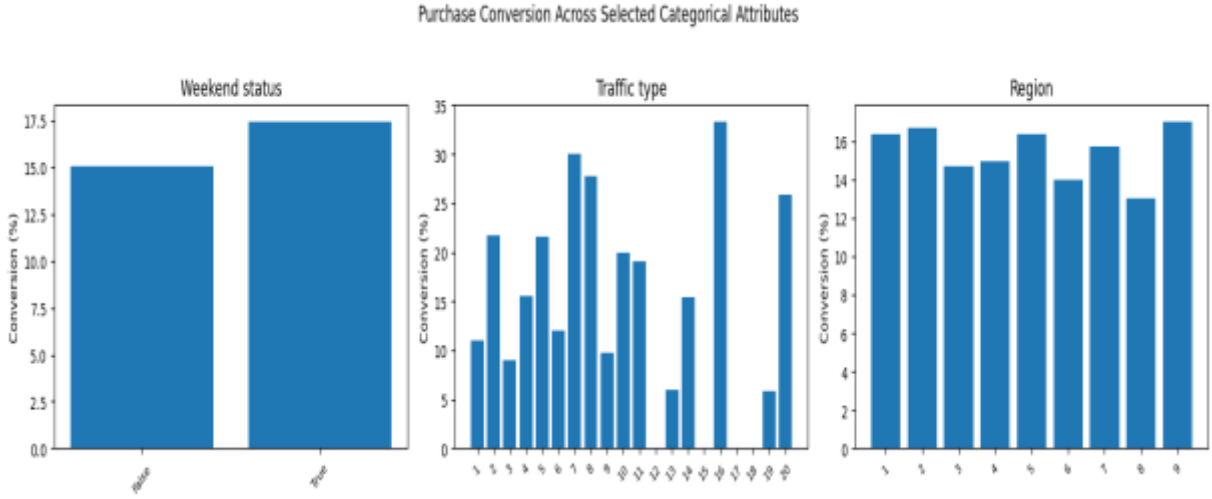



*Figure 5. Conversion by weekend, traffic type and region*

Interpretation: Weekend sessions converted at 17.45% compared with 15.08% on weekdays. Traffic Type 2 converted at 21.66% (847 of 3,911), while Traffic Type 13 converted at 5.91% (43 of 728). Region 9 had the highest regional rate (17.03%) and Region 8 the lowest (12.99%). Small traffic-type groups have unstable rates and should be treated cautiously.

### 11.3 Multivariate analysis

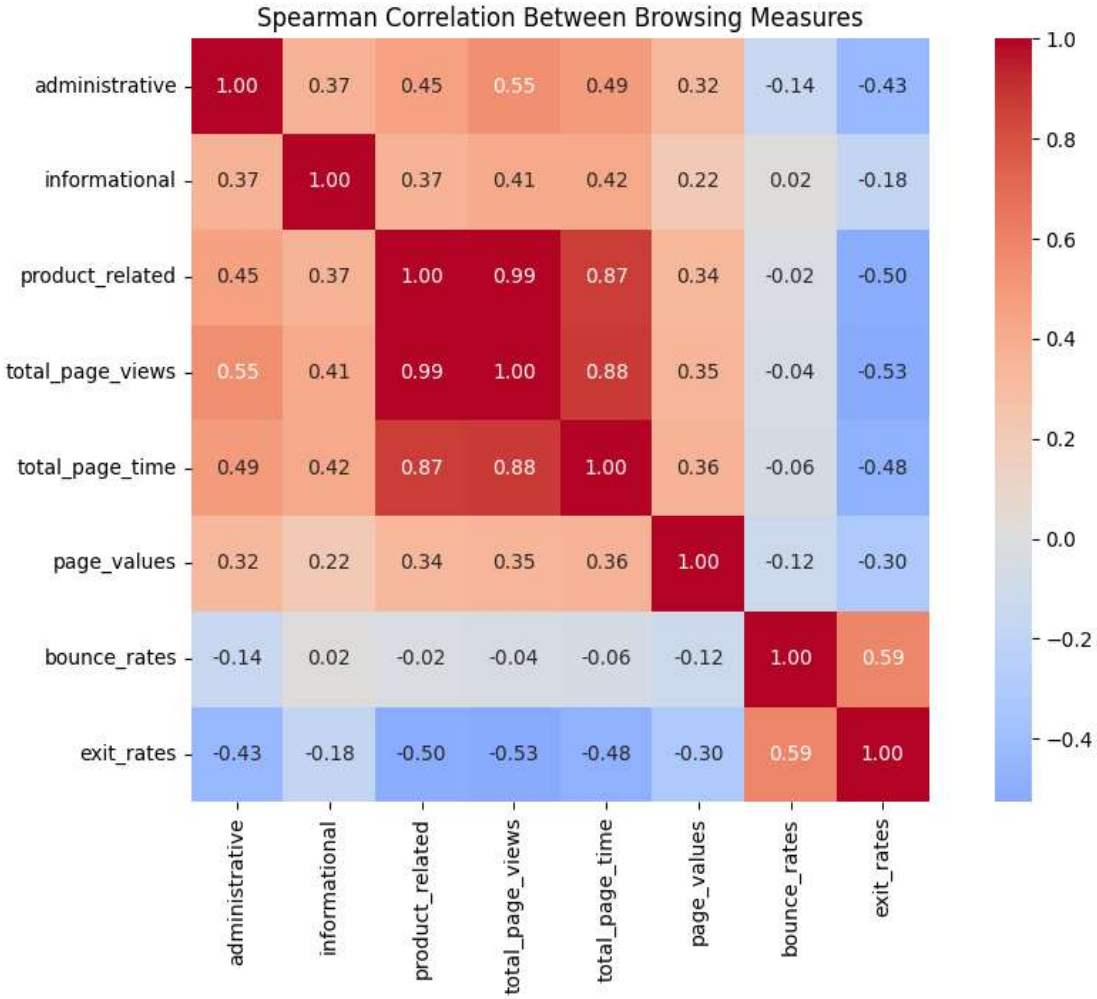



*Figure 6. Spearman correlation heatmap of browsing measures*

Interpretation: Total page views and total page time had a strong positive correlation (0.88). Product-related page views and total page views had a correlation of 0.99; product-related page views and product-related time had a correlation of 0.87. Bounce rate and exit rate had a correlation of 0.59. These are associations, not causal relationships. The very high relationship with total page views is partly expected because the total includes the page-category counts.

The notebook also compares page-value groups: Zero page value – 9,475 sessions, 370 purchases, 3.91% conversion; Low (0–5] – 458 sessions, 162 purchases, 35.37%; Medium (5–20] – 1,102 sessions, 527 purchases, 47.82%; High (>20) – 1,170 sessions, 849 purchases, 72.56%.

## 12. Summary of Findings

1. Overall session conversion was 15.63%: 1,908 purchases from 12,205 cleaned sessions.
2. Purchasing sessions had a median of 32 total page views, compared with 18 for non-purchasing sessions.
3. Purchasing sessions had a median total page time of 1,252.08 seconds, compared with 600.75 seconds for non-purchasing sessions.
4. Purchasing sessions had a median of 29 product-related page views and 1,109.91 seconds on product pages; non-purchasing sessions had 16 views and 526.00 seconds.
5. Median PageValues was 16.76 for purchasing sessions and 0 for non-purchasing sessions.
6. Conversion by page-value group ranged from 3.91% for zero page value to 72.56% for high page value (>20).
7. New visitors converted at 24.93%; returning visitors converted at 14.09% and represented 10,431 sessions.
8. November conversion was 25.49%; February conversion was 1.66%.
9. Weekend conversion was 17.45%, compared with 15.08% on weekdays.
10. Traffic Type 2 had a 21.66% conversion rate, while Traffic Type 13 had 5.91%; regional rates ranged from 12.99% in Region 8 to 17.03% in Region 9.

## 13. Types of Analysis

### 13.1 Descriptive analysis

Used to summarize historical session data through counts, proportions, mean, median, standard deviation, minimum and maximum values, and conversion-rate tables.

### 13.2 Diagnostic analysis

The notebook explores possible reasons for differences in purchase outcomes by comparing page engagement, visitor type, month, weekend status, traffic type, region and page-value groups. These comparisons identify associations and investigation areas; they do not establish causal explanations.

### 13.3 Predictive analysis

Predictive modelling is not performed in the executed notebook. No machine-learning model, prediction score or forecast is reported.

### 13.4 Prescriptive analysis

Recommendations are proposed from the observed patterns: improve product-page information, investigate non-converting journeys, review returning-visitor experience, and audit lower-converting traffic segments. These actions require validation through further analysis or controlled experiments.

## 14. Future Enhancements and Suggestions

- Build a conversion funnel using landing, product, cart and checkout events to locate specific drop-off stages.
- Add campaign identifiers and traffic-source labels so coded traffic categories can be interpreted meaningfully.
- Add device and product/category information to investigate differences in browsing and conversion.
- Validate monthly patterns with campaign calendars, promotions and additional time periods.
- Run controlled A/B tests for product-page improvements, cart recovery and repeat-visitor experiences.
- Develop an interactive dashboard to monitor sessions, purchases, conversion rate, engagement and abandonment.
- Explore a machine-learning classification model in a separate future phase, with train/test separation and appropriate evaluation metrics.
- Automate notebook execution and periodic reporting.

## 15. Conclusion

The project analyzed 12,205 unique online shopping sessions after removing 125 duplicate records from the original 12,330-row dataset. The session conversion rate was 15.63%. Purchasing sessions showed higher median browsing depth and product-related browsing time than non-purchasing sessions. Conversion also differed across visitor types, months, weekend status, traffic types, regions and page-value groups.

The results provide exploratory evidence that browsing engagement and session characteristics are associated with purchase outcomes. They can guide further investigation into product discovery, repeat-visitor journeys and traffic quality. The notebook is descriptive and exploratory: it does not establish causation or train a predictive model.

### Project Output

The notebook exports an analysis ready file named online_shoppers_cleaned_engineered.csv containing the cleaned data and engineered features.

## References

Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository. https://doi.org/10.24432/C5F88Y.

UCI dataset page:
https://archive.ics.uci.edu/dataset/468/online%2Bshoppers%2Bpurchasing%2Bintention%2Bdatase

Project dataset repository: https://github.com/Bhavya-Chellapandian/online-shoppers-purchasing-intention-dataset